



Predicting Customer Churn in Retail Banking

Finding customers who are about to leave, before they go

28,382 CUSTOMERS · 18.53% CHURN · CLASSIFICATION

The Churn Challenge in Retail Banking

Savings accounts fund the bank. Customers often drain balances and go dormant without formally closing, so the bank discovers churn after the money has gone.



The Scale

28,382 customers
21 features
18.53% churn rate



The Aim

Classify every customer as **high risk** or low risk so the team contacts only those likely to leave.



Success Metrics

- **Recall ≥ 0.70**
- **ROC-AUC ≥ 0.80**
- **Accuracy NOT USED**

The Data Landscape

Source: Kaggle Bank Churn Data
28,382 rows × 21 columns

DATA QUALITY

- 0 DUPLICATE ROWS
- 3,871 MISSING VALUES
- HIGH SKEW (5.9M MAX)

RETAIL BANKING CHURN

Customer

6 fields: age, gender, occupation, dependents, tenure.

Location

City (1,605) and Branch (3,185) location codes.

Balance

6 fields across current/prior month and two quarters.

Activity

4 transaction fields: money in and out (monthly).

Recency

Date of last transaction recorded for each customer.

Target

Binary: 1 = Churned, 0 = Stayed.

Churn Is a Two-Month Drain

Churn is not sudden. It is a visible, predictable depletion of funds over approximately 60 days.

Actionable Insight: The bank must act when withdrawal spikes occur, rather than waiting for the balance to hit zero.

STAYED VS. CHURNED

Money Out (This Month)

Stayed: 22 **Churned: 1,120**

51x higher activity

Current Balance

Stayed: 3,643 **Churned: 1,541**

Collapsed to half

Self-Employed: 19.84% Churn

1-3 Dependents:

IDENTIFY THE SPIKE

The Data Was Hiding Its Own Signal

THREE PROBLEMS THE DATA HID

Found	Detail
3,223 Nulls	Dates stored as text "NaT"
806 Ages	Minimum age was 1 year old
10 Deps	Values up to 52; gap from 9 to 25

Informative missingness: No transaction date = **9.28%** churn vs **19.72%** with one.

RETAIL BANKING CHURN

1

Skew Correction

Skew up to **143** was breaking raw correlations.
Log transformation restored the signal.

2

Correlation Boost

Feature	Raw	Log
Current Debit	0.048	0.236

3

Categorical Power

Chi-Square: All categorical features significant ($p < 0.05$), with **Occupation** strongest.

ZERO ROWS DELETED DURING
CLEANING

Model the Change, Change, Not Just Just the Level

The strategy converts raw banking snapshots into orchestrated features that capture the **dynamics of churn**.

ENGINEERED ONCE,
VALIDATED CONTINUOUSLY.

RETAIL BANKING CHURN

Engineering

5 new features: balance_drop_pct, net_flow, debit_to_balance.

Selection

Dropped 4 highly correlated balance columns; 24 final features.

Preprocessing

Yeo-Johnson for skew, Frequency encoding for high-cardinality locations.

Imbalance

Compared SMOTE vs. **class_weight='balanced'** (4.4x penalty).

Splitting

75/25 Train-Test split with **stratified sampling** to preserve churn ratio.

Models

6 tuned runs across Logistic Regression, Random Forest, and XGBoost.

Model 1: Logistic Regression

GridSearchCV scored on recall, StratifiedKFold cv=5.

Best: C=0.01, penalty='l1', class_weight='balanced'

L1 regularisation shrank 10 of the 24 features to zero, allowing the model to perform its own feature selection. Churners caught rose from 553 to 947 out of 1,315.

Performance vs Base

METRIC	BASE	TUNED	CHANGE
Recall	0.421	0.720	+0.299
ROC-AUC	0.815	0.815	-
Precision	0.708	0.486	-0.222
F1	0.528	0.580	+0.052

RECALL TARGET MET

L1 FEATURE SELECTION

Key Coefficients

FEATURE	COEFFICIENT	IMPACT
balance_drop_pct	-1.009	Strong Negative
debit_to_balance	+0.305	Positive
no_last_txn	-0.255	Negative
prev_month_debit	+0.139	Positive

BALANCED WEIGHTS

Random Forest Balances Recall and Precision

Best: depth 6, leaf 20, 200 trees, balanced weights.

GridSearchCV scored on recall, StratifiedKfold cv=5

The Random Forest model provides a stable balance between catching churners and maintaining list quality, heavily leveraging engineered signals.

Performance vs Base Model

METRIC	BASE	TUNED	CHANGE
Recall	0.592	0.703	+0.111
ROC-AUC	0.834	0.839	+0.005
Precision	0.638	0.512	-0.126

Feature Importances

FEATURE	IMP.	ORIGIN
balance_drop_pct	0.393	Engineered
current_balance	0.181	Original
balance_change	0.103	Engineered

39% DRIVEN BY 1 FEATURE

4 OF TOP 5 ENGINEERED

STABLE ROC-AUC RANKING

Model 3: XGBoost Delivers Best Ranking

Best: 200 estimators, depth 3, scale_pos_weight 4.40.
GridSearchCV scored on recall, StratifiedKfold cv=5

XGBoost achieved the highest ROC-AUC of all runs.
By using scale_pos_weight, it effectively handles the 18.5% churn imbalance to prioritize recall.

Performance vs Base Model

METRIC	BASE	TUNED	CHANGE
Recall	0.624	0.700	+0.076
ROC-AUC	0.816	0.842	+0.026
Precision	0.558	0.516	-0.042

HIGHEST ROC-AUC (0.842)

ENGINEERED FEATURE DOMINANCE

Feature Importances

FEATURE	IMPORTANCE	RANK
balance_drop_pct	0.498	#1
no_last_txn	0.062	#2
balance_change	0.039	#3

OPTIMAL BUSINESS TRADE-OFF

XGBoost Is the Best Business Trade-off

All three models met the core targets: **Recall ≥ 0.70** and **ROC-AUC ≥ 0.80** .

Why XGBoost? It wins 4 of 5 metrics. While Logistic Regression has 0.020 higher recall, it results in 139 more wasted calls—a poor trade for the bank.

MODEL PERFORMANCE MATRIX

MODEL	REC.	PREC.	AUC
Baseline	0.000	0.000	0.500
LogReg	0.720	0.486	0.815
Random Forest	0.703	0.512	0.839
XGBoost	0.700	0.516	0.842

Efficiency Gain

XGBoost identifies **920** churners with only **863** wasted calls, maximizing retention ROI.

TARGETS MET: 0.70+ / 0.80+

Business Impact & Next Steps

From zero to seven in ten. The baseline found 0 churners; the final model finds about **7 in every 10**.

LIMITATIONS

Snapshot history, unknown churn rule, precision ~0.52.
Improve with service records and live testing.

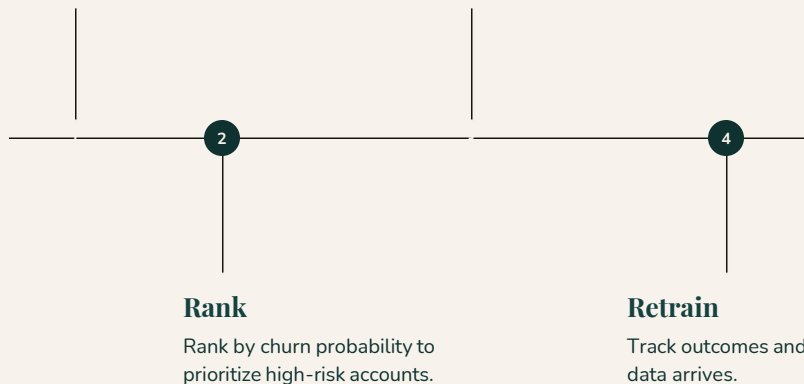
RETAIL BANKING CHURN

Score

Score every customer monthly using the XGBoost pipeline.

Contact

Send top 25% (1,783 accounts) to the retention team.



CONCLUSION



Thank You



Questions & Discussion

XGBOOST · RECALL 0.700 · ROC-AUC 0.842 · BOTH TARGETS MET

Retail
Banking

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